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Details of Project Group:

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Title of the Project: CareerVault: A Web 2.5 Career Verification Platform

a) Objectives: List of objectives with details

- **To develop a Web 2.5 Hybrid Verification Platform:** To build a system that utilizes a traditional PostgreSQL database for fast, low-cost user interactions while leveraging the Polygon blockchain strictly as a decentralized trust anchor.
- **To enable "Zero-Friction" Document Issuance:** To provide organizations with a "DocuSign-like" experience for issuing career documents (LORs, Experience Letters), completely abstracting away the complexities of cryptographic keys and digital wallets.
- **To ensure Data Privacy and GDPR Compliance:** To implement a cryptographic "Salting" mechanism where an off-chain salt is combined with document data before hashing. This ensures that user data can be mathematically "destroyed" upon request (Right to be Forgotten) without breaking the immutable blockchain.
- **To automate Semantic Skill Extraction (AI/NLP):** To implement a Natural Language Processing (NLP) pipeline that parses unstructured text in recommendation drafts and extracts core professional competencies, embedding them as structured data into a W3C-compliant JSON-LD document.
- **To implement a Verified Talent Matchmaking Engine:** To build a search and ranking algorithm that allows recruiters to discover candidates based exclusively on cryptographically verified skills, ranking them by skill match percentage and document recency.
- **To ensure Algorithmic Fairness via Explainable AI (XAI):** To implement transparency mechanisms within the Talent Matchmaking engine, allowing recruiters to see the exact feature weights (e.g., skill match, recency) that contributed to a candidate's rank, thereby eliminating "black-box" bias and ensuring compliance with emerging AI hiring laws.

b) Requirement analysis: requirement analysis document

 Career Verification Platform SRS

c) Dataset : details of the dataset required for the application along with its sources: NA

- d) **Methodology / Algorithm:** Methodology suggested for the application with proper justification

1. The Hybrid Web 2.5 Architecture (Off-chain Processing)

- **Methodology:** All user interactions, authentication (Magic Links via JWT), and document drafting happen on a traditional NestJS/PostgreSQL stack.
- **Justification:** Public blockchains are too slow and expensive to handle standard application state changes (CRUD operations). By keeping the operational layer off-chain, the system achieves enterprise-grade speed and UX, matching current Web2 standards.

2. The "Anchor Engine" & Merkle Tree Batching (On-chain Security)

- **Methodology:** Instead of writing every document to the blockchain directly, the system calculates a SHA-256 hash of each JSON-LD document combined with a random salt. Every 24 hours (00:00 UTC), a background Cron Job collects all new hashes, constructs a binary Merkle Tree, and writes only the single Root Hash to a Polygon Smart Contract.
- **Justification:** This algorithmic approach solves the "gas fee" bottleneck. It allows the system to anchor thousands of documents for the cost of a single transaction (~\$0.01). Furthermore, anyone can independently verify a document using its specific Merkle Proof without needing to access the entire database.

3. NLP Skill Extraction & Matchmaking Algorithm (The Smart Layer)

- **Methodology:** When a manager drafts a letter, the text is passed through an NLP microservice (e.g., SpaCy). Named Entity Recognition (NER) identifies skills and maps them to a standard ontology. These skills are embedded in the JSON-LD before hashing. Later, a ranking algorithm sorts candidates by comparing requested recruiter skills against the verified JSON-LD arrays, applying weights for recency.
- **Justification:** A blockchain only guarantees that data wasn't tampered with; it doesn't guarantee the data is useful. By algorithmically extracting and standardizing skills *before* they are anchored, the platform upgrades from a static "document storage" system to an intelligent, cryptographically backed talent search engine.

4. Explainable AI (XAI) for Algorithmic Transparency

- **Methodology:** The matchmaking ranking is augmented with an Explainable AI layer, specifically utilizing SHAP (SHapley Additive exPlanations) values or interpretable decision trees. When the system ranks Candidate A over Candidate B, the XAI layer extracts the marginal contribution of each variable (e.g., presence of a specific verified skill, recency of the document) to the final score.
- **Justification:** In recruitment technology, AI bias is a severe legal and ethical liability. Standard neural networks act as "black boxes." By utilizing XAI, the platform not only matches candidates using cryptographically verified data but also provides mathematically sound, human-readable justifications for every recommendation. This bridges the gap between zero-trust blockchain data and trustworthy AI decision-making.



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Remark of Guide:

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